Spectral Parameters Selection Based on Multiagent Deep Reinforcement Learning for AOTF Infrared Multispectral Imaging System

Na Li, *Member, IEEE*, Xiangyu Yang, Huijie Zhao, *Member, IEEE*, and Xiaoyu Wang

***Abstract*—The infrared multispectral imaging system based on an acousto-optic tunable filter (AOTF) possesses strong capabilities in suppressing complex backgrounds and anti-interferences for the detection of aerial targets. It is worth noting that the wavelengths and bandwidths of the detection bands have a significant impact on detection capability. However, the existing spectral parameters selection methods have poor generalization, rely on manual reasoning strategies and repeated training with a large amount of data, and cannot achieve the optimum for the spectral parameters. To address these problems, we propose the wavelength and bandwidth selection models based on multiagent reinforcement deep learning (WBMARL). This is the first attempt to apply the multiagent reinforcement learning to spectral parameters selection tasks, which provides a new paradigm. Based on the assumption that the combinatorial optimization problem of spectral parameters selection is regarded as a Markov game process, the WBMARL, which consists of a deep feature extraction module and two agents, is designed. Through cooperation among the agents, the Nash equilibrium point is reached, and the optimal spectral parameters selection policy is obtained. The experimental results on the infrared multispectral images captured by an AOTF system indicate that the spectral parameters selected by WBMARL are more conducive to the detection of targets in complex backgrounds and anti-interference.**

***Index Terms*—Acousto-optic tunable filter (AOTF) system, infrared multispectral imaging detection, multiagent deep reinforcement learning, spectral parameters selection.**

I. Introduction

THE infrared imaging detection is an important approach to aerial target detection and precision guidance. The current infrared panchromatic/dual-band imaging systems have many successful applications in this field, but also face the following two challenges. First, clouds, buildings, and other complex backgrounds with complex structures or strong radiation make the detection process produce a large number of false alarms [1]. Second, various types of infrared interferences confuse the detection systems by strong radiation or similar radiation and motion characteristics with the target, making it difficult for the detection systems to accurately distinguish between the target and interference [2], [3]. In this case, infrared multispectral imaging detection has become a feasible solution, which can effectively suppress the complex background and reduce the false alarm rate by using the appropriate detection bands. It can distinguish between the target and the interference according to the difference in spectra, which improves the ability of anti-interference.

There are many types of spectral imaging systems, among them the infrared multispectral imaging system based on acousto-optic tunable filter (AOTF, AOTF system), with the advantages of gaze imaging, tunable detection bands, and fast tuning speed, is suitable for the detection of high-speed aerial moving target [4]. The AOTF system operates based on the acousto-optic effect. When a radio frequency (RF) signal is applied to the piezoelectric transducer, it generates ultrasonic waves within the acousto-optic medium, thereby causing diffraction of light at the corresponding wavelength. By changing the RF signal, the light of different wavelengths can be selected, realizing the tuning of the bands. The detection bands of the AOTF system can be quickly set up according to the task, then images of different bands can be obtained one by one, and the spectrum of the target can be extracted from the images, as shown in Fig. 1. In addition, according to the bandwidth required by the task, the neighboring bands can be integrated through image processing to obtain synthetic images with corresponding bandwidth, so as to achieve better signal-to-clutter ratio (SCR) and better detection results.

In many cases, there are significant differences between the spectra of target, interference, and background in certain bands, while in others they appear highly similar. Desirable detection results require bands with high SCR and significant spectral differences. The wavelengths of the selected bands influence both the SCR and spectral differences, as do the bandwidths. A narrow bandwidth may lead to a lower SCR, while an excessive bandwidth could reduce spectral differences, thereby degrading detection performance. Therefore, the selection of spectral parameters for the detection bands, including the wavelength and bandwidth, is essential for the AOTF system [5]. In addition, due to changes in observation conditions, the spectra of targets, interferences, and back-

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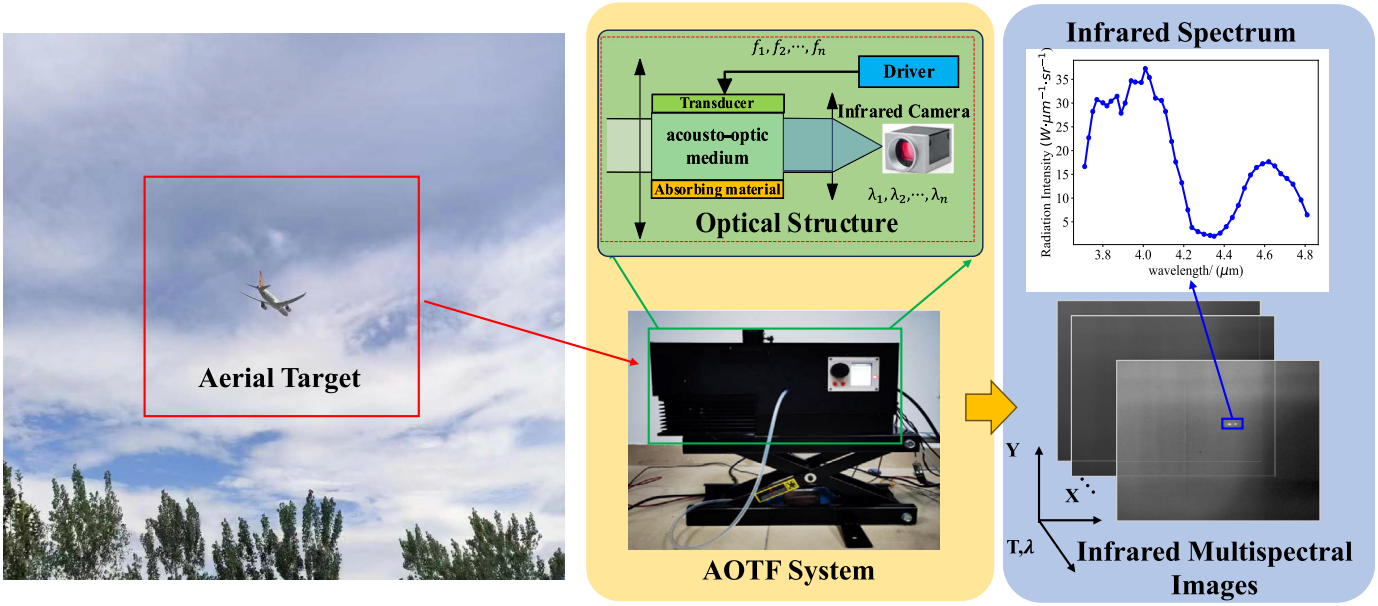


Fig. 1. Optical system of the AOTF system mainly consists of lens, AOTF, and infrared camera. By changing the RF signals, the AOTF system can obtain spectral images one by one, from which the spectrum of the target can be obtained.

grounds may undergo significant variations, also available data is often expensive and limited, which requires spectral parameters selection model to exhibit good generalization performance under data-constrained conditions.

For spectral parameters selection, researchers have done a lot of research around selection metrics and selection methods. The selection metrics are mainly based on information, distance, correlation, and contrast. The selection methods include traditional band selection methods and intelligent methods based on intelligent search, deep learning, and reinforcement learning. Among these methods, the methods based on intelligent search can obtain the global optimum, yet they have poor generalization ability and rely heavily on manually reasoning strategies. Deep learning methods have good generalization and excellent feature extraction capabilities, but they depend on repeated training with a large amount of data. To some extent, selection methods based on reinforcement learning addresses these challenges by requiring relatively minimal training data while maintaining strong generalization performance. These reinforcement learning methods have the structure of a single agent, and only select discrete wavelengths. The design of the reward serves hyperspectral classification, and the solution method generally adopts deep Q-learning suitable for discrete action spaces. However, the design is not suitable for the spectral parameters selection of the AOTF system. First, in terms of the agent system design, current methods all adopt a single-agent system, which lacks the ability to select both wavelength and bandwidth. If discrete wavelengths and continuous bandwidths are designed as the action of a single agent, it may lead to the curse of dimensionality. It is worth noting that the influence of wavelength and bandwidth on the detection effect is coupled. The wavelength and bandwidth selection tasks should not be simply divided into the wavelength selection task and the bandwidth selection task independently, as this may prevent achieving the optimum. Second, regarding the reward function design, the reward used for hyperspectral classification cannot be directly applied to selecting bands beneficial for target detection. In addition, the deep Q-learning is not suitable for the selection of a continuous bandwidth.

Therefore, after considering model generalizability, data requirements, and the effectiveness and stability of spectral parameters selection, the multiagent deep reinforcement learning-based model named WBMARL is designed, where the wavelength and bandwidth selection tasks are regarded as Markov game process. According to the requirements of spectral differences between the target and the interference and SCR of the bands, the state, action, reward, state transition function, and discount factor of the multiagent system are defined. The WBMARL comprises a spectral feature extraction module (SFEM) and two agents. The SFEM is used to extract features from spectral images to construct the environment. The two agents are, respectively, used for wavelength and bandwidth selections. Through the cooperation between the agents and their interaction with the environment, the optimal policy for wavelength and bandwidth selections is obtained. In addition, due to the interaction among agents and the environmental nonstationarity, the multiagent deep deterministic policy gradient (MADDPG) algorithm is used to train the model to achieve stable convergence, and the discrete wavelengths are processed for continuous action spaces. The contributions are as follows.

1) This is the first time to solve the wavelength and bandwidth selections by using a multiagent reinforcement learning model to obtain a joint optimal solution, providing a new paradigm for the spectral parameters selection.
2) The reward function is defined according to the spectral differences and SCR of images for target detection.

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3) The WBMARL, composed of an SFEM and two agents, is designed.
4) The WBMARL is trained by using the MADDPG to learn the selection policy.

Section II details the relevant research on spectral parameters selection, Section III introduces our method, Section IV verifies and analyzes our model, and Section V is the conclusion.

II. Related Works

Spectral parameters selection can be regarded as a combinatorial optimization problem, which can be analyzed from two perspectives: objective function formulation and optimization approach, which, respectively, correspond to the selection metrics and selection methods. This section briefly reviews works relevant to spectral parameters selection from the perspectives of both selection metrics and methods.

A. Selection Metrics

The selection metrics are broadly classified into the following four categories: information-based metrics, distance-based metrics, correlation-based metrics, and contrast-based metrics.

1) Information-Based Metrics: The information-based metrics whose aim is to select the subset of bands with the richest information, which usually involve metrics such as information entropy, mutual information [6], [7], and eigenvalues of the covariance matrix [8]. However, these information-based metrics are very sensitive to noise with low robustness and reliability.

2) Distance-Based Metrics: Distance-based metrics seek to select bands that contribute to spectral classification with different distance metrics, such as Bhattacharyya Distance and Jeffreys–Matusita (JM) Distance [9], [10].

3) Correlation-Based Metrics: Correlation-based metrics, which select less correlated bands to reduce information redundancy, usually use metrics such as spectral correlation coefficient, spectral information scatter, and orthogonal projection scatter [11].

4) Contrast-Based Metrics: The metrics based on the contrast or difference between target and background are mainly used to select the bands which is conducive to target detection, such as the SCR [12], signal-to-noise ratio (SNR) [13], and the spectral contrast [14].

Above metrics provide valuable support for spectral parameters selection in various application scenarios. In practical applications, the combination of multiple metrics is more effective [15], such as optimal index factor (OIF) [16]. The following diverse spectral parameters selection methods essentially employ combinations of these metrics as objective functions to pursue optimal solutions.

B. Selection Methods

From the perspective of the selection methods, which include traditional band selection methods such as those based on classical search, ranking, clustering, low-rank representation, transform domain, and intelligent band selection methods based on intelligent search, deep learning, and reinforcement learning.

1) Traditional Band Selection: The traditional band selection methods have been widely applied in practice.

a) Search-based methods: Selection methods based on classical search depend on the way of gradually adding or removing bands to find the optimal band subset. Its core is to optimize the object function, such as improving classification accuracy, and reducing the correlation between bands and maximizing information [17].

b) Ranking-based methods: Selection methods based on ranking quantify the importance of bands according to the evaluation metrics, and selects bands based on a specific threshold or the number of bands. Xu et al. [18] introduced structural similarity to measure the relationships between the bands and proposed a similarity-based ranking strategy.

c) Clustering-based methods: Selection methods based on clustering divides bands into several groups and selects the band from each group to form the final combination [19], [20]. The selection methods based on K-means clustering [21], nearest neighbor clustering [22], and affinity propagation clustering are typical representatives. Miao et al. [23] proposed a band selection method which is carried out based on the structural similarity among bands. Then, in each group, the bands with large information entropy and low noise are selected to form a subset.

d) Representation-based methods: Selection methods based on low-rank representation makes use of the prior information that the affinity matrix obtained from the low-rank model can directly reflect the contribution of each band. Yu et al. [24] put forward the a low-rank representation model with super-pixel and imbalanced classwise constraints. Chang et al. [25] proposed a band selection method based on self-representation model that fully takes both the spatial and spectral structure of the original hyperspectral cube into consideration.

e) Transform-based methods: Selection methods based on transform domain treat the spectrum as a signal for transformation, including Fourier transform, wavelet transform, and so on, and then select characteristic points, such as local extreme values. Bruni et al. [26] presented an adaptive wavelet-based band selection method that selects relevant bands by analyzing spectral signatures of classes of interest.

f) Subspace-based methods: Selection methods based on subspace also have extensive applications. They seek the subspaces that are most beneficial for downstream tasks in the feature space composed of bands, including subspace partitioning [27], [28] and subspace projection [29].

Traditional methods have been extensively studied and widely applied, but they have poor generalization to different data, there is still room for improvement in their performance.

2) Intelligent Band Selection: The intelligent band selection methods can be divided into three categories, including those based on intelligent search, deep learning, and reinforcement learning.

a) Intelligent search-based methods: Intelligent search algorithms are mainly heuristic search methods, including physics-based methods, evolutionary-based methods [30], and swarm intelligence-based methods. Zhang et al. [31] used an adaptive multistrategy swarm optimization and dynamic

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adjustment of update strategies for band selection. The ant colony optimization is applied to select a small-band subset such that classification accuracy can be maintained or even improved [32]. Yong et al. [33] used an improved artificial bee colony algorithm for band selection of hyperspectral images. In conclusion, the heuristic search methods can often obtain solutions close to the global optimum. However, they heavily rely on the reasoning rules designed based on expert knowledge, and seriously lack generalization.

b) Deep learning-based methods: Deep learning that can automatically extract features, avoiding manual feature design, has become a hot trend in spectral data processing and has shown effectiveness in the problem of band selection. Deep learning-based band selection methods frequently employ a data reconstruction strategy [34] and an attention mechanism [35]. Wang et al. [36] adopted a self-supervised architecture and designed an auxiliary task for hyperspectral data reconstruction. Zhao et al. [37] designed a band selection method based on the self-attention mechanism and spectra reconstruction, and utilized the attention matrix to obtain the importance of different bands. Furthermore, deep learning methods integrated with graph learning have also been extensively applied [38], [39], and the hybrid graph convolutional network is employed for feature extraction and refinement [40]. In addition, some of the special models and strategy designs are highly innovative. Yu et al. [41] adopted the principle of sparsity to select bands through a deep learning network. Sun et al. [42] introduced a domain adaptation strategy to enhance generalization capability. Yang et al. [43] proposed utilizing LiDAR data to guide hyperspectral band selection in multimodal fusion applications. Deep learning methods are also used for feature extraction and combined with other methods, such as heuristic search [44]. Although the band selection methods based on deep learning have excellent effects and a certain improvement in generalization, they rely on repeated training with a large number of samples, which is unacceptable in detection scenarios with insufficient data.

c) Reinforcement learning-based methods: The selection methods based on reinforcement learning are precisely designed to solve the problems caused by artificial design of reasoning rules, insufficient generalization, and repeated training with a large number of samples. Reinforcement learning obtains policies that maximize rewards or achieve specific goals through the interaction and trial-and-error of agents with the environment. Mou et al. [45] mapped band selection to a Markov decision process (MDP) and proposed a solution method based on deep Q-learning. It is the first study that makes use of deep reinforcement learning for the task of band selection. Feng et al. [46] first applied semisupervised learning and reinforcement learning to the band selection task. Tisha et al. [47] also modelled band selection as MDP and solved it using deep Q-learning. In addition, reinforcement learning methods can also be combined with heuristic search to integrate the advantages of both [48]. The above methods demonstrate the effectiveness of reinforcement learning methods in the band selection task. They do not rely on large amounts of training data and demonstrate strong generalization performance. However, there are still some issues. First, it is very important that these methods are only used to select wavelengths and cannot simultaneously select wavelengths and bandwidths. Second, these methods construct reward using information entropy, mutual information, correlation, etc., which fail to characterize the difference between the target and the background. Third, the above methods employ deep Q-learning. However, deep Q-learning is not suitable for continuous action spaces, which conflicts with the continuous-valued bandwidths.

Therefore, the SCR and spectral similarity are used to construct the reward of the WBMARL, and a multiagent system consisting of two agents is developed to select wavelengths and bandwidths. These two agents sequentially select the corresponding wavelengths and bandwidths, and they collaborate to obtain the optimal policy. In addition, considering the interaction and cooperation between the two agents, as well as the nonstationary environment issues that may arise due to state changes, the MADDPG algorithm is chosen. Moreover, the discrete wavelengths are processed to meet MADDPG's requirements for continuous action spaces.

III. Methodology

A. Problem Formulation

The spectral parameters selection task for the AOTF system can be modeled as a constrained multiobjective combinatorial optimization problem with two key characteristics. First, there is a coupling relationship between wavelength and bandwidth as variables. Second, the objective function consists of multiple components. Taking spectral angle (SA) and SCR as examples, the former is used to maximize interclass differences, while the latter is used to maximize intraclass SNR. These principles are fundamental criteria in the target detection process. For a multispectral image H obtained, with the n bands, the image size $h \times w$, the wavelengths of the n bands being $\lambda_H = \{\lambda_1, \lambda_2, \ldots, \lambda_n\}$, and the bandwidths being $\Delta\lambda_H = \{\Delta\lambda_1, \Delta\lambda_2, \ldots, \Delta\lambda_n\}$, the task is to select K sets of wavelengths and bandwidths based on H to maximize the following objective function, which are denoted as $\{\lambda_{(s_1)}, \lambda_{(s_1)}, \ldots, \lambda_{(s_k)}\}$ and $\{\Delta\lambda_{s_1}, \Delta\lambda_{s_2}, \ldots, \Delta\lambda_{s_k}\}$, respectively,

$$\begin{aligned} \max_{\{(\lambda_i,\Delta\lambda_i)\}_{i=1}^{K}} \quad & J(\lambda,\Delta\lambda) = \alpha_1 \sum \mathrm{SA}(\lambda,\Delta\lambda) \\ & + \alpha_2 \sum \mathrm{SCR}(\lambda,\Delta\lambda) \\ \text{s.t.} \quad & \lambda_i \in \lambda_H \quad \forall i = 1,\ldots,K \\ & \Delta\lambda_i \in [\Delta\lambda_{\min}, \Delta\lambda_{\max}] \\ & K < n. \end{aligned} \tag{1}$$

Here, $\sum \mathrm{SA}(\lambda,\Delta\lambda)$ represents the SA obtained with the selected spectral parameters, $\sum \mathrm{SCR}(\lambda,\Delta\lambda)$ represents the SCR obtained with the selected spectral parameters, and α_1 and α_2 are weighting coefficients used to balance the two terms. $\Delta\lambda_{\min}$ and $\Delta\lambda_{\max}$ are the lower and upper bounds of the bandwidth values, respectively. Theoretically, $\Delta\lambda_{\min}$ corresponds to the bandwidth of the current wavelength, while $\Delta\lambda_{\max}$ corresponds to the bandwidth across multiple adjacent wavelengths.

Based on this problem description, it can be observed that this optimization problem is an NP-hard problem. Two

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sources of complexity can be identified: the curse of dimensionality arising from the nearly infinite combinations formed by discrete wavelengths and continuous bandwidths, and the nonlinear coupling relationship between wavelength and bandwidth selections. Directly applying a single-agent reinforcement learning framework to this problem faces significant challenges, as a single agent must simultaneously reason about the choice of wavelength and bandwidth.

To address the problem, the problem is formulated within a cooperative multiagent deep reinforcement learning paradigm. This approach draws inspiration from distributed optimization and game theory, decomposing the coupled parameters selection into two interconnected subproblems handled by specialized agents, one for wavelength selection and another for bandwidth selection. This decomposition naturally alleviates the curse of dimensionality and allows each agent to develop expertise within its respective action domain.

However, in a cooperative multiagent system, the environment becomes nonstationary from the perspective of any individual agent, as the policy update of one agent alters the state transition dynamics and reward distribution for the others. This nonstationarity violates the fundamental Markov assumption of the standard single-agent reinforcement learning algorithm and can lead to unstable training and failure to converge. Therefore, the MADDPG algorithm is adopted, which employs a centralized training with a decentralized execution paradigm. During training, a centralized critic network guides the individual actors toward cooperative policies. This framework is theoretically grounded to drive the multiagent system toward a Nash equilibrium, representing a stable and jointly optimal policy for the coupled spectral parameters selection task.

To ensure the stability and convergence of the multiagent system, several conditions must be satisfied, including the boundedness and Lipschitz continuity of the reward function, the compactness of the state–action space, and the Robbins–Monro condition for the learning rate. The design of WBMARL is precisely based on these convergence requirements and the characteristics of the spectral parameter selection task. Sections III-B and III-C detail the formalization of this Markov game process and the instantiation of the WBMARL model within this theoretical framework.

B. Markov Game Process

The spectral parameters selection task can be regarded as a multiagent sequential decision problem and is modeled as a Markov game process. Based on this assumption, we propose WBMARL, where multiagent system including two agents is designed, one agent selects the wavelength, while the other selects the bandwidth. The two agents have a cooperative relationship, and their state updates exhibit the Markov property. Based on their joint decision, the synthesized images are generated. To achieve optimal detection performance, the reward is constructed using SCR and SA, both of which are influenced by the chosen wavelengths and bandwidths. In addition, the wavelength and bandwidth exhibit mutual dependence, meaning their selections must be jointly optimized for the best outcome. The environment of the multiagent

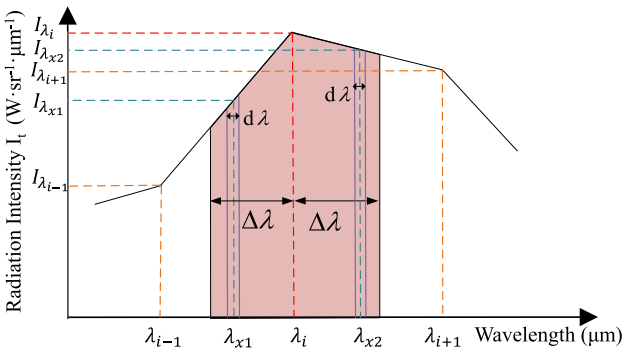


Fig. 2. Synthesized image I_t is obtained by linear interpolation and integration, vertical axis indicates radiation intensity.

system is designed based on the Markov game process and can be represented by a tuple $(S, A^1, A^2, T, R^1, R^2, \gamma)$, where S represents the state of the multiagent system, A^1 and A^2 represent the action of the wavelength selection agent and the bandwidth selection agent, respectively, T represents the state transition function, R^1 and R^2 represent the rewards of the wavelength selection agent and the bandwidth selection agent, respectively, and γ represents the discount factor.

Furthermore, the synthesized images can be obtained based on the spectral parameters selection result. If the selected wavelength is λ_i and the corresponding selected bandwidth is $2\Delta\lambda$, then the corresponding starting wavelength and ending wavelength are λ_{x_1} and λ_{x_2}. For a given pixel, the radiation intensity across its different spectral bands can be denoted as $I(\lambda_{i-1}), I(\lambda_i)$, and $I(\lambda_{i+1})$ in the data H, the radiance intensity $I(\lambda_{x_1})$ and $I(\lambda_{x_2})$ at λ_{x_1} and λ_{x_2} are obtained by linear interpolation

$$I\left(\lambda_{x_1}\right) = I\left(\lambda_{i-1}\right) + \frac{I\left(\lambda_i\right) - I\left(\lambda_{i-1}\right)}{\lambda_i - \lambda_{i-1}} \times \left(\lambda_{x_1} - \lambda_{i-1}\right) \tag{2}$$

$$I\left(\lambda_{x_2}\right) = I\left(\lambda_i\right) + \frac{I\left(\lambda_{i+1}\right) - I\left(\lambda_i\right)}{\lambda_{i+1} - \lambda_i} \times \left(\lambda_{x_2} - \lambda_i\right). \tag{3}$$

The radiance intensity I_t of the synthesized image within the range of the bandwidth $2\Delta\lambda$ is calculated by integration, as shown in Fig. 2

$$I_t = \int_{\lambda_i - \Delta\lambda}^{\lambda_i} I\left(\lambda_{x_1}\right) d\lambda + \int_{\lambda_i}^{\lambda_i + \Delta\lambda} I\left(\lambda_{x_2}\right) d\lambda. \tag{4}$$

The environmental model of the multiagent system is modeled as follows.

1) State: The state S_t of the multiagent system at time step t is constructed from the spectral images H. As shown in Fig. 3, rather than using raw high-dimensional data directly which is redundant and can obscure task-relevant information, the SFEM based on a pretrained ResNet-101 model is employed. This module processes each spectral band individually to extract high-level spatial–spectral features, which are then projected into a compact feature vector via a multilayer perceptron. $S_t = [F_s, lw_t]$, where F_s represents the deep features extracted from the H, and lw_t represents the currently selected spectral parameters. This design is motivated by the need to provide a compact, informative, and task-relevant

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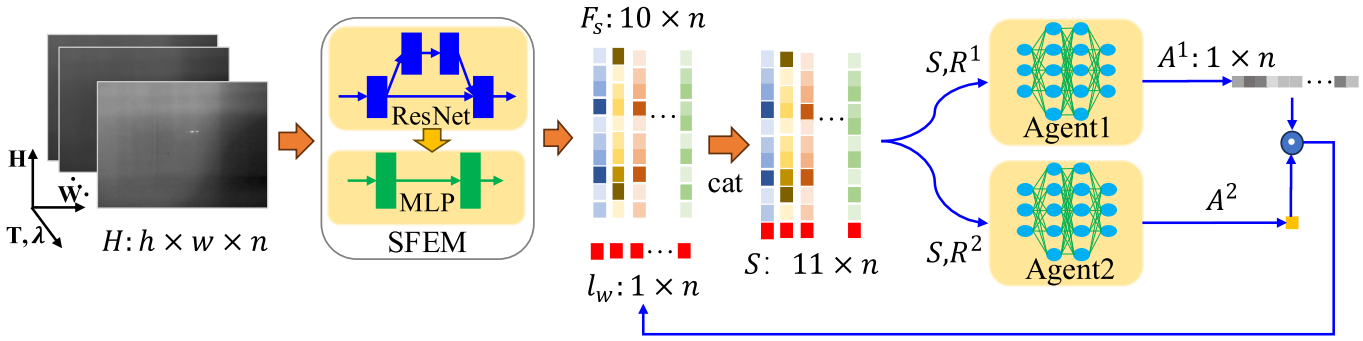


Fig. 3. WBMARL composed of an SFEM and two agents is designed. The SFEM is used to extract features from spectral images, and the two agents are used for wavelength and bandwidth selections. The environment model of the multiagent system that includes state, action, reward, transition function, and discount factor is defined based on the Markov game process.

state representation for the agents. By mapping the raw high-dimensional spectral images onto a lower-dimensional manifold that captures essential features for target detection, the curse of dimensionality in the state space is effectively mitigated. This alleviates the learning difficulty for the agents and lays the foundation for efficient policy learning. After the feature extraction of H, the spectral features F_s with a size of $n \times 10$ are obtained. The lw_t is a vector with a length of n, representing the selected wavelengths and bandwidths, which is obtained by the product of the actions of two agents. Both of F_s and lw_t jointly constitute the state with a size of $n \times 11$. F_s is a normalized feature with well-defined upper and lower bounds. lw_t is multiplied by the already selected spectral parameters, where both bandwidth and wavelength are bounded. Therefore, lw_t also possesses clear upper and lower bounds, ensuring that the designed state space is a compact set.

2) Action: The action A^1 of the wavelength selection agent is represented as a vector with a length of n, whose elements range from 0 to 1. If the value corresponding to the index i in A^1 is the largest, it means that the λ_i is selected from λ_H, and the sum of the A^1 is 1. Through this processing, the action space for wavelength selection is continuous. The action A^2 of the bandwidth selection agent is a scalar, representing the selected bandwidth. If the wavelength selection agent currently selects the λ_i, and the bandwidths of $[\lambda_{i-2}, \lambda_{i-1}, \lambda_i, \lambda_{i+1}, \lambda_{i+2}]$ are $[\Delta\lambda_{i-2}, \Delta\lambda_{i-1}, \Delta\lambda_i, \Delta\lambda_{i+1}, \Delta\lambda_{i+2}]$, respectively, then the minimum value of the current action A_t^2 of the bandwidth selection agent is $\Delta\lambda_i$, and the maximum value is $\sum_{i-2}^{i+2} \Delta\lambda_m$. The design of actions ensures that the action space is a compact set, which is a necessary condition for guaranteeing the convergence of the model.

3) Reward: For target detection, two fundamental principles are to maximize interclass differences and intraclass SNR. To serve this purpose, it is required that the SCR of the selected bands be high to facilitate target detection in different scenarios, and that the spectral differences between the target and the interference in the selected bands be large to facilitate the distinction. Therefore, the rewards of the agents are set according to the SCR and the spectral similarity, the former should be as large as possible, while the latter should be as small as possible. It is important to note that these two rewards are coupled in their influence on the selection of wavelength and bandwidth. For instance, increasing the bandwidth may lead to an improvement in SCR while simultaneously reducing spectral differences. Therefore, the two agents must collaborate to balance these potentially conflicting objectives.

The reward R^1 of the wavelength selection agent is represented by the SA between the target and the interference or between the target and the nearby background in all the selected bands. The reward of the wavelength selection agent is defined as $R_t^1 = \alpha_1 \mathrm{SA}(S_t, A_t^1, A_t^2)$, where α_1 is a trainable parameter and has a value range of 0–1. The SA between the target spectrum S_{target} and the spectrum S_{other} of the interference or background in synthesized images is expressed as follows:

$$\mathrm{SA}\left(S_t, A_t^1, A_t^2\right) = \arccos\left(\frac{S_{\text{target}} \cdot S_{\text{other}}}{|S_{\text{target}}| \cdot |S_{\text{other}}|}\right). \tag{5}$$

The reward R^2 of the bandwidth selection agent is represented by the SCR of the synthesized images corresponding to the selected wavelength λ_i according to A_t^1 and the selected bandwidth according to A_t^2 at the current step t. The radiance intensity in the synthesized image is denoted as I_t. The reward is defined as $R_t^2 = \alpha_2 \mathrm{SCR}(A_t^1, A_t^2)$, where α_2 is a trainable parameter and has value range of 0–1

$$\mathrm{SCR}\left(A_t^1, A_t^2\right) = \frac{\left|\mu_{\text{target}} - \mu_{\text{background}}\right|}{\sigma_{\text{background}}}. \tag{6}$$

Among them, μ_{target} is the mean value of the target with size of $ks \times ks$ in the synthesized image, $\mu_{\text{background}}$ is the mean value of the background with size of $3ks \times 3ks$ surrounding the target, and $\sigma_{\text{background}}$ is the standard deviation of the background surrounding the target. The reward of the multiagent system is obtained by combining the two agents, that is $R_t = R_t^1 + R_t^2$.

The SA ranges between 0 and pi, while SCR does not tend to infinity when the gray value is bounded, and its variance is nonzero. In addition, trainable weight parameters within the range of 0–1 are introduced, ensuring that the reward function is bounded for any state–action pair (S_t, A_t^1, A_t^2). Furthermore, based on the definitions of SA and SCR, SA is constructed from the arccosine function, which satisfies Lipschitz continuity, and SCR also satisfies Lipschitz continuity when the gray

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value is bounded and its variance is nonzero. Consequently, the reward function also exhibits Lipschitz continuity. For any state–action pair (S_t, A_t^1, A_t^2), there exists a constant L_R that satisfies the following relationship:

$$\begin{aligned}&\left|R_t\left(S_t, A_t^1, A_t^2\right) - R_{t+1}\left(S_{t+1}, A_{t+1}^1, A_{t+1}^2\right)\right| \\ &\quad \le L_R \left\|\left(S_t, A_t^1, A_t^2\right) - \left(S_{t+1}, A_{t+1}^1, A_{t+1}^2\right)\right\|. \end{aligned} \tag{7}$$

The reward function ensures continuous and bounded properties that are crucial for model convergence.

4) State Transition: It represents the probability distribution P that maps the actions A_t^1 and A_t^2 taken by the multiagent system in the current state S_t to the next state S_{t+1}. The agents generate A_t^1 and A_t^2 based on the S_t, while the next state S_{t+1} is determined by the A_t^1, A_t^2, and S_t. Consequently, the next state S_{t+1} depends solely on the current state S_t, satisfying the Markov property. The state transition function is deterministic and is represented as follows:

$$P\left(S_{t+1}\middle|S_t, A_t^1, A_t^2\right) = \begin{cases} lw_{t+1} = lw_t \text{if } JM_{t+1} < JM_t \\ lw_{t+1} = A_t^2 \cdot A_t^1 + lw_t \text{ else.} \end{cases} \tag{8}$$

Among them, JM_t represents the JM distance between spectra of the target and the background, as well as between the spectra of the target and interference in the synthesized images according to the selected spectral parameters. When the JM distance begins to decrease, it means that the newly added band has led to a decline in separability, and the band selection has reached the optimum

$$\mathrm{JM} = 2\left(1 - e^{-B}\right) \tag{9}$$

$$\begin{aligned} B = &\frac{1}{8}(\mu_1 - \mu_2)^T \left(\frac{\Sigma_1 + \Sigma_2}{2}\right)^{-1} (\mu_1 - \mu_2) \\ &+ \frac{1}{2} \ln \frac{\left|\frac{\Sigma_1 + \Sigma_2}{2}\right|}{\sqrt{|\Sigma_1||\Sigma_2|}}. \end{aligned} \tag{10}$$

Here, μ_1 and μ_2 are the mean values of the spectra, and Σ_1 and Σ_2 are the covariances, respectively.

5) Discount Factor: $\gamma \in [0, 1)$is used to adjust the influence of long-term rewards in reinforcement learning, and its value ranges between 0 and 1.

C. Wavelength and Bandwidth Selection Models Based on Multiagent Reinforcement Learning

The WBMARL consists of SFEM and two agents. The SFEM is used to extract features from multispectral images and construct the environment. The two agents, which are based on the actor–critic structure, are responsible for wavelength and bandwidth selections. Through the cooperation between the agents as well as their interaction with the environment, the optimal policy for wavelength and bandwidth selections can be obtained.

The spectral feature extraction of the multispectral images is to compress and abstract the data captured by the AOTF system into a more compact and advanced latent form, thereby capturing the important features of the different bands. ResNet101 is used to conduct deep feature extraction band by band. The deep features extracted from n bands are utilized to construct the state space, and the pretrained model parameters are adopted. ResNet101 can be used to obtain the deep feature with a size of $[n, c, h, w]$. Then, MLP is used to project the spectral features into a vector F_s

$$F_s = \mathrm{MLP}\left(\mathrm{ResNet}\left(H\right)\right). \tag{11}$$

The agents adopt the actor–critic structure. Both the wavelength selection agent and the bandwidth selection agent contain one actor network and one critic network. This architecture implements a centralized training with a decentralized execution paradigm, which is crucial for effective cooperation in multiagent system. The actor network generates actions based on the state of the multiagent system, and it is trained to execute in a decentralized manner, ensuring operational efficiency during deployment. The critic network can observe the global state and the actions of all agents. This centralized perspective enables the critic to evaluate the joint action, guiding the actors toward policies that maximize the collective reward of the team. This mechanism is central to addressing the nonstationarity of the environment caused by concurrent policy updates of multiple agents.

As shown in Fig. 4, for the actor network of the wavelength selection agent, its input is the flattened state S_t of the multiagent system, and its output is an $n+1$-D vector composed of the action A_t^1 and the coefficient α_1. The actor network consists of two fully connected layers. For the critic network of the wavelength selection agent, its input is the state and actions (S_t, A_t^1, A_t^2) of the multiagent system, and its output is the evaluation Q_1 of the action A_t^1. The critic network is also composed of two fully connected layers.

For the actor network of the bandwidth selection agent, its input is the flattened state S_t of the multiagent system, and its output is a 2-D vector composed of the action A_t^2 of the bandwidth selection agent and the coefficient α_2. The actor network consists of two fully connected layers. For the critic network of the bandwidth selection agent, its input is the state and actions (S_t, A_t^1, A_t^2) of the multiagent system, and its output is the evaluation Q_2 of the action generated by the actor network of the bandwidth selection agent. The critic network is also composed of two fully connected layers.

D. MADDPG Algorithm

Since WBMARL is a small-scale multiagent system with cooperative relationships between agents and a continuous action space, MADDPG is selected to address the environmental nonstationarity caused by agent actions. The MADDPG algorithm is adopted to train WBMARL, using the paradigm of centralized training with decentralized execution to alleviate the problem of nonstationary training environment and promote stable convergence. It has been previously established that the reward function is bounded and satisfies Lipschitz continuity, and that both the state space and action space are compact sets. During the training process, by employing an appropriate decaying learning rate η_t to satisfy the Robbins–Monro condition

$$\sum_{t=0}^{\infty} \eta_t = \infty, \quad \sum_{t=0}^{\infty} \eta_t^2 < \infty. \tag{12}$$

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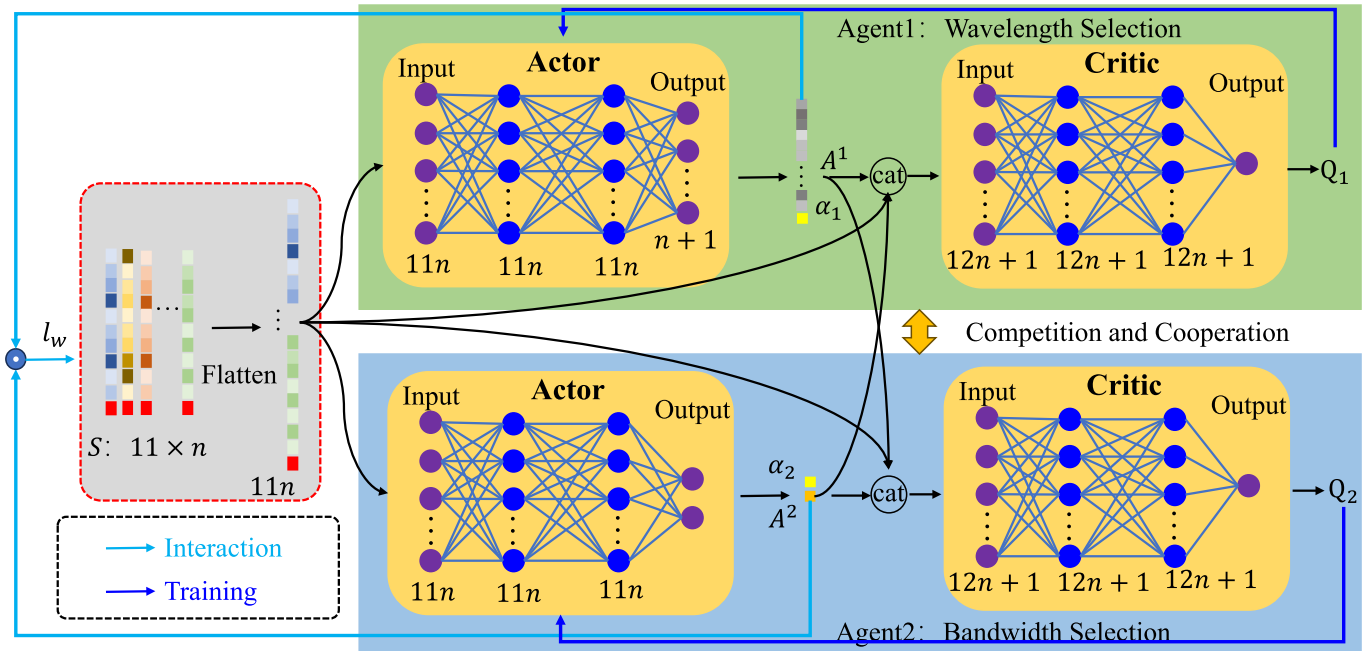


Fig. 4. Agents network comprises two agents with actor–critic structure. The actor networks take actions under the current state according to their own policy, while the critic networks can observe the global states and actions.

Thus, the boundedness and Lipschitz continuity of the reward function, the compactness of the state and action spaces, and the satisfaction of the Robbins–Monro condition for the learning rate collectively guarantee the model's convergence and stability. In the multiagent system, the goal of the agents is to maximize the expected return $J(\theta_1, \theta_2, S)$. The policies of the agents are represented as $A_t^1 = \mu_1(S_t)$ and $A_t^2 = \mu_2(S_t)$, the parameters of the actor networks of are represented as θ_1 and θ_2

$$J(\theta_1, \theta_2, S) = \mathbb{E}_{\mu_1,\mu_2}\left[\sum_{t=0}^{\infty} \gamma^t R_t \,\middle|\, S_0 = S, a_t^1 \sim \mu_1, a_t^2 \sim \mu_2\right]. \tag{13}$$

After training, the multiagent system can reach a Nash equilibrium. For the optimal strategy pair μ_1^*, μ_2^* at a Nash equilibrium, as follows, no agent can improve a value function starting from any initial state by unilaterally deviating from this strategy:

$$\begin{aligned} J^{(\mu_1^*,\mu_2^*)}(S) &\geq J^{(\mu_1,\mu_2^*)}(S) \quad \forall\mu_1 \ \forall S \\ J^{(\mu_1^*,\mu_2^*)}(S) &\geq J^{(\mu_1^*,\mu_2)}(S) \quad \forall\mu_2 \ \forall S. \end{aligned} \tag{14}$$

During the training process, the experience replay mechanism is adopted to reuse the training samples. After the initialization and update of the agents, experiences can be generated during the execution process, which can be represented as $(S_t, A_t^1, A_t^2, R_t, S_{t+1})$, and these experiences are stored in the experience pool M. During training, experiences are randomly sampled in batches from the experience pool for training, so as to reuse the past experiences and break the correlation of the sequential experiences, maintaining the stable convergence of the training process.

The target network mechanism is used to achieve the stable convergence of the network. The policies of the current network of the agents are represented as $A_t^1 = \mu_1(S_t)$ and $A_t^2 = \mu_2(S_t)$, the parameters of the actor networks of the current network are represented as θ_1 and θ_2, and the parameters of the critic networks of the current network are represented as π_1 and π_2. The policies of the target network are represented as $A_t^{1,'} = \mu_1'(S_t)$ and $A_t^{2,'} = \mu_2'(S_t)$, the parameters of the actor networks of the target network are represented as θ_1' and θ_2', and the parameters of the critic networks of the target network are represented as π_1' and π_2'. Initially, the parameters of the target network are the same as those of the current network. During the training process, the critic network is updated first. N samples $(S_t^j, A_t^{1,j}, A_t^{2,j}, R_t^j, S_{t+1}^j) \sim M$ are randomly sampled in batches from the experience pool, where $j = 1, 2, \ldots, N$. The loss function of the critic network is defined as follows:

$$L_i = \frac{1}{N}\sum_{j=1}^{N}\left(y^{i,j} - Q_i^{\mu_i}\left(S_t^j, A_t^{1,j}, A_t^{2,j}\right)\right)^2 \tag{15}$$

$$y^{(i,j)} = R_t^j + \gamma Q_i^{\mu_i'}\left(S_{t+1}^j, A_{t+1}^{1,j,'}, A_{t+1}^{2,j,'}\right). \tag{16}$$

Among them, $i = 1,2$, and $Q_i^{\mu_i}(S_t^j, A_t^{1,j}, A_t^{2,j})$ is the Q value output by the critic of the current network, and $Q_i^{\mu_i'}(S_{t+1}^j, A_{t+1}^{1,j,'}, A_{t+1}^{2,j,'})$ is the Q value output by the critic of the target network. The critic network, π_1 and π_2, is updated by minimizing the loss function. Then, the actor network is updated using the policy gradient $\nabla_{\theta_i} J$

$$\nabla_{\theta_i} J \approx \frac{1}{N}\sum_{j=1}^{N}\nabla_{\theta_i}\mu_i\left(S_t^j\right)\nabla_{A_i^j} Q_i^{\mu_i}\left(S_t^j, A_t^{1,j}, A_t^{2,j}\right). \tag{17}$$

Among them, $i = 1, 2, \nabla_{\theta_i}\mu_i(S_t^j)$ and $\nabla_{A_i^j} Q_i^{\mu_i}(S_t^j, A_t^{1,j}, A_t^{2,j})$ are the gradients calculated by the actor network and the

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critic network, respectively. After updating the current network multiple times, the target network is then updated, which can be represented as follows:

$$\theta_i' \leftarrow \tau\theta_i + (1-\tau)\,\theta_i', \quad i = 1,2 \tag{18}$$

$$\pi_i' \leftarrow \tau\pi_i + (1-\tau)\,\pi_i', \quad i = 1,2 \tag{19}$$

where $0 < \tau < 1$, τ is used to adjust the learning rate. In addition, in order to prevent the training process from falling into local optimal solutions and ensure the balance between exploration and exploitation of the agents, the ϵ-greedy strategy is adopted to select actions at each time step. Here, $0 < \epsilon < 1$, the best action is taken according to the policy that the agents have already learned with a probability of $1-\epsilon$, and an action is randomly selected with a probability of ϵ.

Algorithm 1 MADDPG Algorithm

Randomly initialize actors $\theta_1, \theta_2, \theta_1', \theta_2'$
Randomly initialize critics $\pi_1, \pi_2, \pi_1', \pi_2'$
Initialize experience pool M
For $e = 1 \rightarrow$ Epochs **do**:
  Select actions A_t^i with ϵ-greedy strategy
  Store transition in M, $(S_t, A_t^1, A_t^2, R_t, S_{t+1}) \rightarrow M$
  For $t = 1 \rightarrow$ Steps **do**:
    Sample batch from M
    Compute target Q-values
    Update critics via L_i minimization
    Update actors via policy gradient $\nabla_{\theta_i} J$
    Soft update target networks $\theta_i' \leftarrow \tau\theta_i + (1-\tau)\theta_i'$, $\pi_i' \leftarrow \tau\pi_i + (1-\tau)\pi_i'$, $i = 1,2$
  End For
End For

The training procedure of WBMARL is shown in the MADDPG algorithm. By decomposing the complex parameters joint optimization problem and utilizing a multiagent system, the curse of dimensionality is effectively mitigated. The SA and SCR rewards are directly aligned with the core objectives of the detection task, endowing the learning process with clear interpretability and task-specific adaptability. Built upon the MADDPG framework, the centralized critic coordinates the agents, ensuring training stability and convergence in nonstationary environments, ultimately reaching an equilibrium point through interagent collaboration.

IV. Experiments and Analysis

A. Dataset Description

The spectral images captured by the AOTF system are used to verify the effectiveness of WBMARL. The wavelength of the AOTF system is continuously adjustable within the range of 3.7–4.8 μm with a bandwidth of 31–52 nm, and the image size is 256×320. The central wavelengths and bandwidths of different bands are shown in Fig. 5. The gray images are converted into radiation intensity images after radiation calibration. There are two datasets, namely Dataset 1, which includes simulated aerial targets and artificial interferences in different complex backgrounds, and Dataset 2, which includes

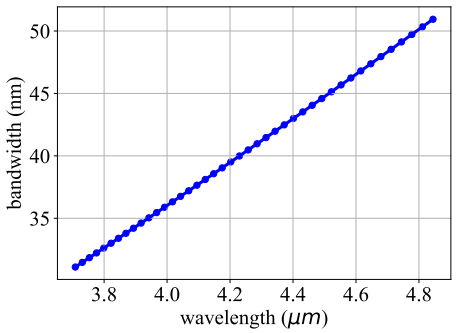


Fig. 5. Bandwidths corresponding to different wavelengths are different. The wavelength of the AOTF system is continuously adjustable within the range of 3.7–4.8 μm with a bandwidth of 31–52 nm.

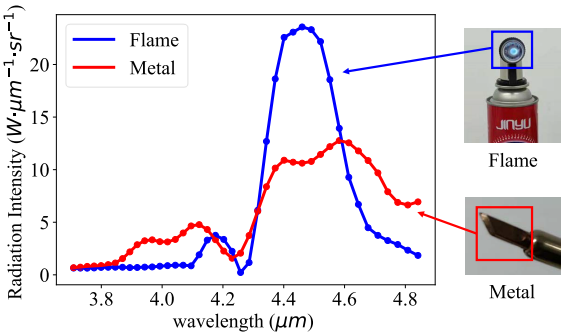


Fig. 6. Spectrum of flame and high-temperature metal in Dataset 1.

the airplane target in a cloud background. In the experiment, first, the data of 42 bands are captured, the training and testing of the spectral parameters selection methods are carried out based on these data, the result analysis of parameter selection is the same as well. Then, according to the selected spectral parameters, the spectral images are selected to verify the effectiveness through target detection.

1) Dataset 1: In which the flame of butane combustion is used to simulate the plume of the airplane, with a temperature of around 800–1000 K, serving as the detection target. High-temperature metal with a temperature of around 700–800 K is used to simulate artificial interferences. The size occupied by the target and the interference did not exceed 9×9. The spectra of the flame and the high-temperature metal are shown in Fig. 6. It can be seen that the spectrum of the flame has an obvious double-peak feature near 4.2 μm, which is the characteristic of the emitted radiation of high-temperature carbon dioxide gas after atmospheric attenuation. The spectrum of the airplane plume is similar, also exhibiting a dual-peak feature. The high-temperature metal has a spectrum similar to the gray body. The total radiation intensity of the target and the interference in the range of 3.7–4.8 μm is close, but their spectra are different. The backgrounds included vegetation background, building background, road background, and sky background. The visible images and the infrared multispectral images of the backgrounds are shown in Fig. 7.

2) Dataset 2: The target is an airplane with a cloud background. When detected from behind the airplane, the

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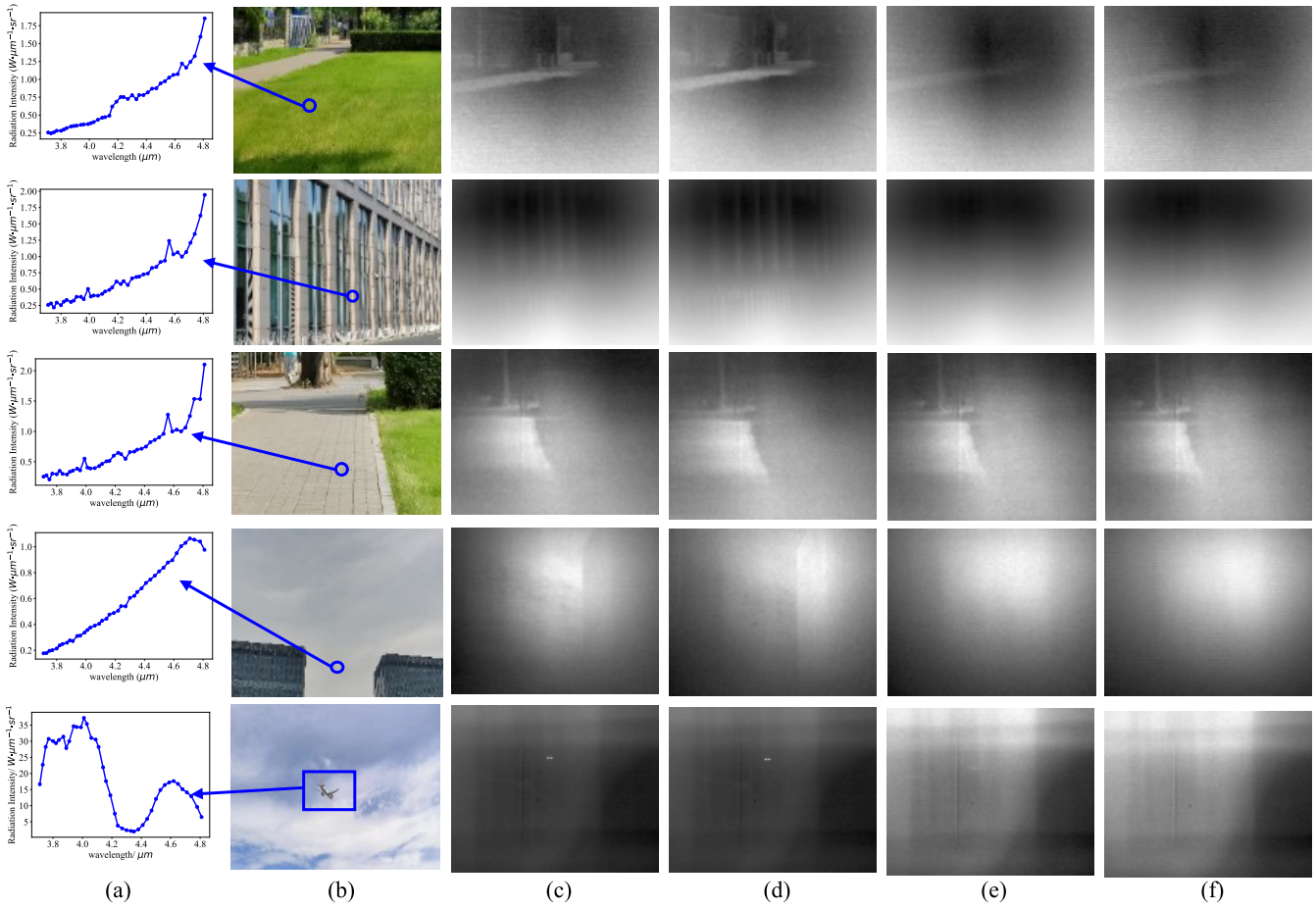


Fig. 7. Spectra, visible images, and infrared multispectral images (a) and (b) spectra and visible images, and (c)–(f) infrared spectral images at 3.71, 3.89, 4.31, and 4.81 μm. (Top four rows) The data with different backgrounds in Dataset 1, which are vegetation background, building background, road background, and sky background, respectively. (Bottom row) The data of the airplane target in Dataset 2.

most prominent radiation source is the exhaust nozzle of the airplane, exhibiting certain gray body radiation. The spectrum of the airplane is shown as Fig. 7. For the band selection method requiring training, the training set includes no fewer than ten multispectral image samples (42 bands) per background, totaling 100 sets. After the parameters selection, 1000 sets of data according to the selection results are selected for the training and testing of the detection model, which is used to validate the effectiveness of the selected bands for target detection. The ratio of the training set to the testing set is 7:3.

B. Experiment Setting

1) Evaluation Metrics: The SA is used to evaluate the difference between the target spectrum and the interference spectrum. The average SCR of all selected images is used to quantitatively evaluate the target detection ability. To quantitatively evaluate the information of the synthesized images according to selected parameters, the mean information entropy (MIE) is used. For each of the images, $p(x)$ denotes the frequency of radiation intensity x, and MIE is the mean value of the information entropy for K images

$$\text{MIE} = \frac{1}{K}\sum_{k=1}^{K}\int p(x)\log_2 p(x)\,dx. \tag{20}$$

Furthermore, in order to verify whether the selected images are conducive to target detection, the detection probability (Pd) and the false alarm rate (Fa) are used, where Pd equals recall rate (Rec), and Fa equals the 1 − Pre, Pre denotes the precision rate

$$P_d = \text{Rec} \tag{21}$$

$$F_a = 1 - \text{Pre}. \tag{22}$$

The calculations for Rec and Pre are shown below, where TP, FN, and FP indicate true positive, false negative, and false positive, respectively,

$$\text{Rec} = \frac{\text{TP}}{\text{FN} + \text{TP}} \tag{23}$$

$$\text{Pre} = \frac{\text{TP}}{\text{FP} + \text{TP}}. \tag{24}$$

In addition, the JM distance is used to characterize the separability of the spectra, which is used to analyze the sensitivity of WBMARL to the number of selected parameters.

2) Comparison Methods: To validate the effectiveness of WBMARL, we employed multiple comparative methods, such as traditional methods, intelligent search-based methods, deep learning-based methods, and reinforcement learning-based methods. Due to the lack of methods capable of simultaneous

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wavelength and bandwidth selections, we augmented certain methods with a bandwidth optimization module based on maximum SCR.

OIF [16]: As a composite metric integrating both image standard deviation and correlation coefficient, this method represents one of the most widely adopted approaches combining multiple metrics. OIF+SCR means first performing the OIF for wavelength selection and then performing bandwidth selection by maximizing the SCR.

ASPS [27]: ASPS dynamically partitions hyperspectral subspaces via adaptive thresholding, optimizing band selection through localized feature importance analysis. ASPS +SCR means first performing the ASPS for wavelength selection and then bandwidth selection by maximizing the SCR.

OPBS [29]: OPBS selects minimally redundant yet discriminative spectral bands by orthogonal projection, effectively maximizing information while eliminating interband correlations. OPBS+SCR means first performing the OPBS for wavelength selection and then bandwidth selection by maximizing the SCR.

AMSPSO [31]: AMSPSO features an adaptive multistrategy particle swarm optimization and dynamic adjustment of particle update strategies for band selection.

CFSP [28]: A subspace partition method using curve fitting based on maximum curvature points and local context representative band selection.

ERAL [48]: An adaptive evolutionary–reinforcement learning algorithm integrates reinforcement learning to choose feasible actions in each iteration, and evolutionary algorithm to update exploration.

S4P [25]: The method utilizes an adaptive and weighted multiple graph fusion term to capture spatial structure in the self-representation space.

CNNeGA [44]: The method integrates a 3-D convolutional neural network as the fitness function in a genetic algorithm.

To verify the detection effectiveness of the selected parameters combinations, the small target detection method based on Bi-Conv-LSTM [49] is adopted to conduct detection on the images selected.

3) Experimental Settings: The operating system of the platform is Windows 10, the CPU is Intel i9, the GPU is RTX4090, and the memory is 32 GB. The experiment is carried out with Python 3.6 and PyTorch. During the training process, the discount factor γ is set to 0.99, the learning rate η is set to 0.0001, the soft update coefficient τ is set to 0.01, the greediness rate ϵ is initially set to 1 and decays to 0.01 with a step size of 0.95, the capacity of the experience pool is 10 000, the capacity of the mini-batch sampling is 100, and the number of training iterations is 1000 to train the spectral parameters selection network. The target detection model is trained and tested on the same platform, and the input time window size of the detection model is the number of combinations selected; other settings are consistent with the Bi-Conv-LSTM [49].

C. Comparison Results

To evaluate the spectral parameters selection of WBMARL, the analysis is conducted on the quantitative description of the spectral parameters selection and the detection effectiveness.

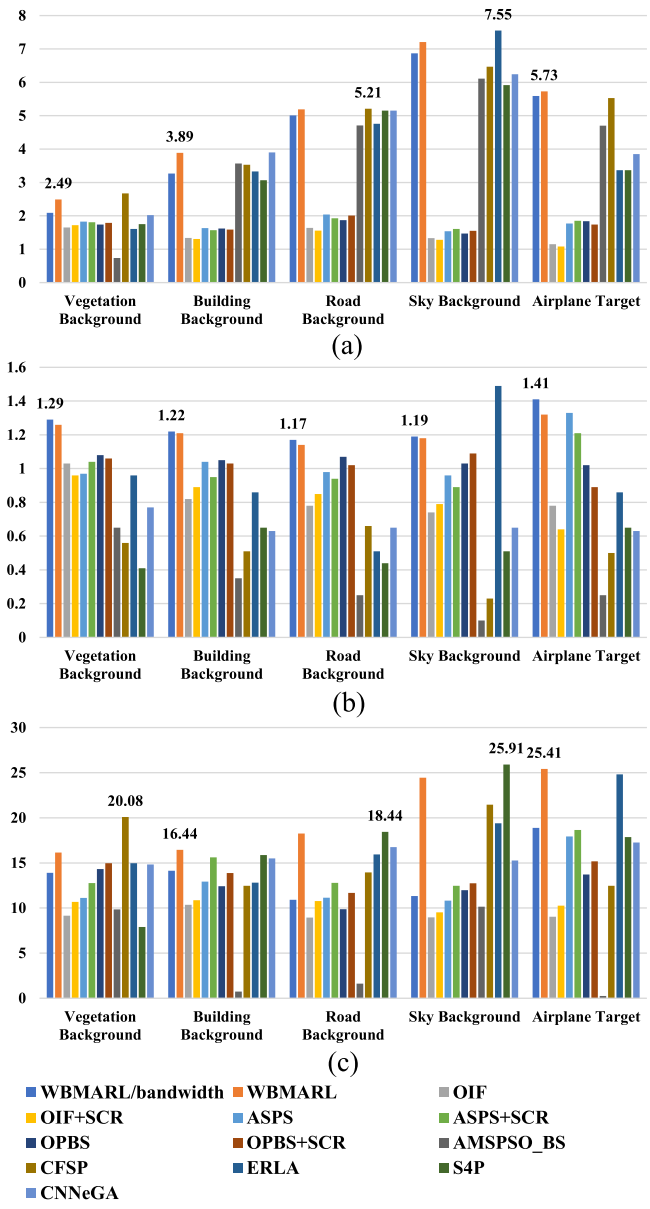


Fig. 8. Evaluation of spectral parameters for Datasets 1 and 2. (a)–(c) Results for MIE, SA, and SCR, respectively. WBMARL/bandwidth indicates that only the wavelength selection results are utilized.

The results of the spectral parameters selection for Dataset 1 are presented in Fig. 8. A comparative analysis reveals that wavelength selection is crucial for the selection of spectral parameters. For WBMARL versus WBMARL/bandwidth, the incorporation of bandwidth demonstrates a negligible impact on MIE, significant SCR improvement, and marginal reduction in spectral difference. This trade-off enhanced SCR at the cost of slightly diminished spectral difference proves acceptable as bandwidth expansion enables adjacent bands integration to boost SCR. Similar patterns are observed in OIF+SCR, ASPS+SCR, and OPBS+SCR comparisons.

Notably, the WBMARL yields images with the largest target–interference spectral difference across all backgrounds and all methods except WBMARL/bandwidth. However, WBMARL does not consistently maximize SCR and MIE in

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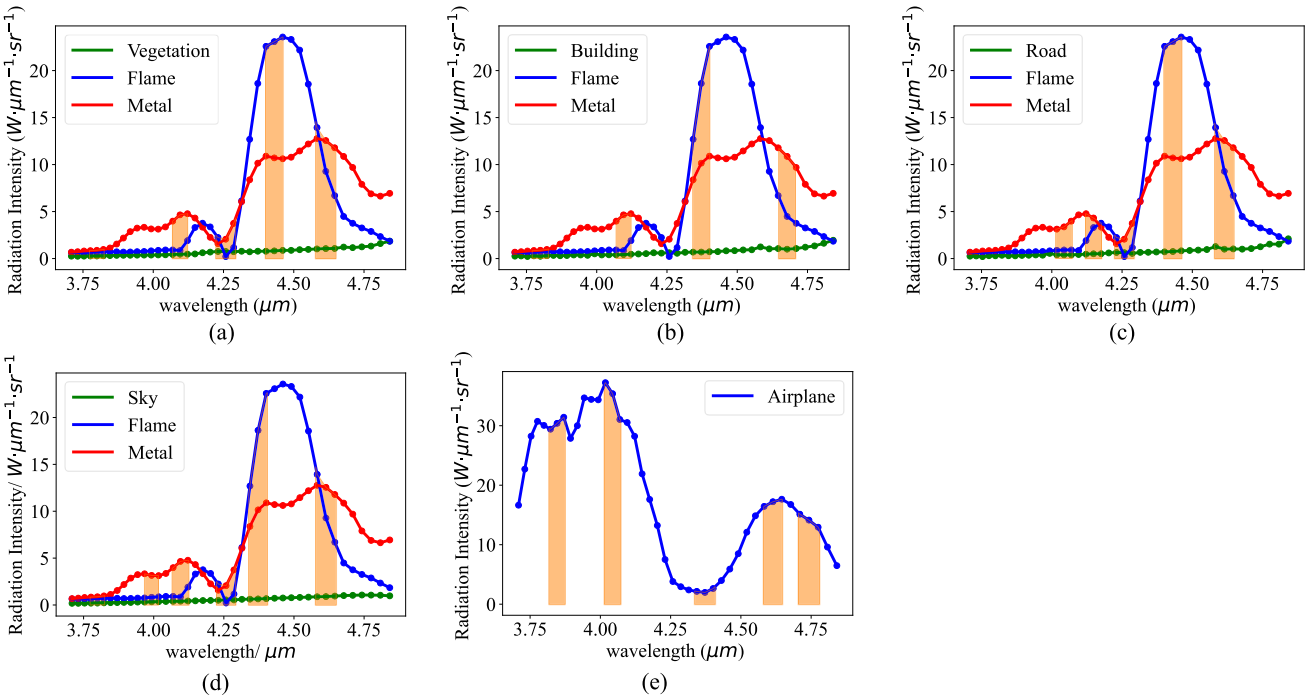


Fig. 9. Spectral parameters selection of WBMARL for Datasets 1 and 2. (a)–(d) Results for vegetation background, building background, road background, and sky background, respectively, in Dataset 1. (e) Result for airplane target in Dataset 2. The orange columnar areas represent the selected spectral parameters, the central position indicates the selected wavelength, and the width represents the bandwidth.

all scenarios. While it achieves the highest MIE for vegetation and building backgrounds, its MIE slightly trails CFSP in road backgrounds and ERLA in sky backgrounds. Similarly, WBMARL only maximizes SCR in building backgrounds, underperforming CFSP and S4P in vegetation, road, and sky conditions. These results demonstrate that WBMARL prioritizes enhancing target–interference discriminability and image quality, which is critical for target detection and multiple targets recognition.

Overall, the WBMARL method achieves a top-tier performance across all three metrics in various backgrounds, demonstrating its exceptional generalization capability and adaptability to diverse scenarios. In comparison, while CFSP and S4P deliver satisfactory results in image quality metrics, such as MIE and SCR, their selected bands exhibit insufficient spectral differences, failing to fully exploit spectral information. ERLA shows balanced performance overall, with its MIE reaching the optimal level only in sky backgrounds, yet still maintaining strong generalization ability. As for the other methods, they either perform well only in specific background types or overemphasize image quality at the expense of spectral differences.

The selection results by WBMARL of Dataset 1 are shown in Fig. 9. Under different backgrounds, the spectral parameter results selected by the WBMARL are not exactly the same. For the distinction between simulated targets and simulated interferences, WBMARL selects five sets of parameters for the vegetation background, four sets of parameters for the building background, five sets of parameters for the road background, and six sets of parameters for the sky background. Moreover, the selected wavelengths are mostly located at positions with high radiation intensity or significant spectral features. At this time, the spectral difference between the target's spectrum and interference's spectrum is relatively significant, indicating that the WBMARL has a certain generalization to different backgrounds.

The results of the spectral parameters selection for Dataset 2 are presented in Fig. 8. WBMARL achieved the highest values in both MIE and SCR, while SA attained the second-highest performance across all metrics except WBMARL/bandwidth. The selected wavelengths and bandwidths for Dataset 2 by WBMARL are shown in Fig. 9.

D. Detection Effectiveness

To quantitatively analyze the detection effectiveness of the selected spectral parameters in different backgrounds, target detection is carried out on the selected images according to selected spectral parameters.

The detection results are shown in Table I. For Dataset 1 across various backgrounds, it can be seen that, compared with other spectral parameters selection methods, the images selected by WBMARL consistently achieved the highest Pd, exceeding 90% in all scenarios. However, WBMARL did not attain the lowest Fa; AMSPSO, CFPS, and S4P achieved minimal Fa values in certain backgrounds. This phenomenon is consistent with the selection pReferences of different methods. While AMSPSO, CFPS, and S4P prioritize bands with optimal image quality, WBMARL sacrifices some SCR to maximize spectral discriminability, consequently resulting in elevated Fa values. The detection results with different backgrounds are shown in Fig. 10, where targets and interferences can be detected and accurately distinguished.

For Dataset 2, compared with other spectral parameters selection methods, the detection according to WBMARL has

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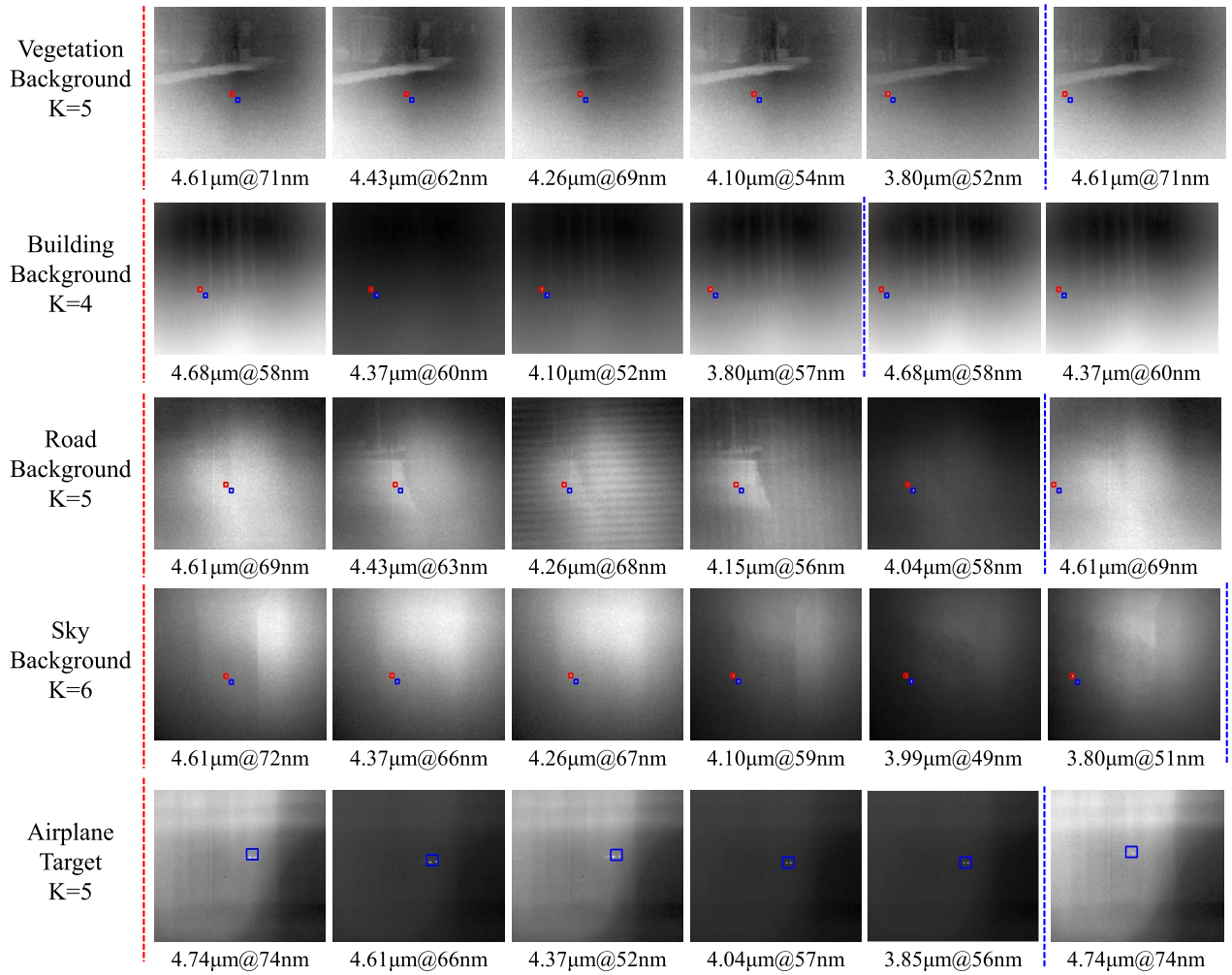


Fig. 10. Results of the detection for Datasets 1 and 2. (Top four rows) The results for vegetation background, building background, road background, and sky background in Dataset 1, respectively. The blue box indicates the position of the target, while the red box indicates the position of the interference. (Bottom row) Results of the detection for airplane target in Dataset 2, where the blue box indicates the position of the target. Acquire data in a loop according to the selection results of spectral parameters. The vertical blue lines represent a new loop.

TABLE I

RESULTS OF THE DETECTION FOR DATASETS 1 AND 2

Backgrounds	Metric	WBMARL	OIF+SCR	ASPS+SCR	OPBS+SCR	AMSPSO	CFSP	ERLA	S4P	CNNeGA
Vegetation Background	Pd	**92.1%**	80.1%	87.1%	83.7%	86.6%	86.1%	88.7%	88.3%	86.4%
	Fa	2.9%	6.2%	4.4%	5.7%	2.0%	**0.1%**	1.6%	1.9%	1.1%
Building Background	Pd	**92.8%**	77.6%	83.1%	85.4%	92.4%	92.3%	88.1%	91.6%	92.1%
	Fa	2.5%	8.3%	6.2%	5.8%	**1.5%**	2.5%	1.5%	3.0%	2.5%
Road Background	Pd	**94.1%**	79.2%	86.8%	84.1%	87.9%	86.9%	80.6%	90.1%	93.9%
	Fa	**1.7%**	6.7%	4.8%	5.3%	6.9%	4.9%	7.8%	6.7%	2.7%
Sky Background	Pd	**90.9%**	73.4%	81.9%	84.1%	90.8%	90.6%	88.8%	89.8%	89.8%
	Fa	3.7%	10.3%	7.3%	6.5%	2.1%	2.5%	7.8%	**1.6%**	2.5%
Airplane Target	Pd	**95.5%**	81.1%	82.3%	87.6%	89.1%	78.2%	87.9%	86.7%	91.7%
	Fa	2.1%	8.4%	7.3%	6.7%	3.9%	1.1%	4.7%	**0.3%**	1.2%

the highest Pd for airplane target. Moreover, the Pd reaches 95.5% that outpaces second-place CNNeGA by 3.8% and the Fa is only 2.1%, demonstrating excellent effectiveness. The detection results for airplane are shown in Fig. 10.

E. Analysis of the Time Complexity

The inference times of different models are shown in Table II. Notably, WBMARL completes spectral parameters selection in 0.713 s and ranks as the third-fastest among all evaluated methods, while the OIF method requires exhaustive computation across all possible band combinations, resulting in significantly higher time consumption. Furthermore, methods including AMSPSO, ERLA, and CNNeGA that require extensive iterative computations for parameter selection or inference exhibit significantly slower processing speeds, reaching hundreds of seconds. In contrast, CFSP,

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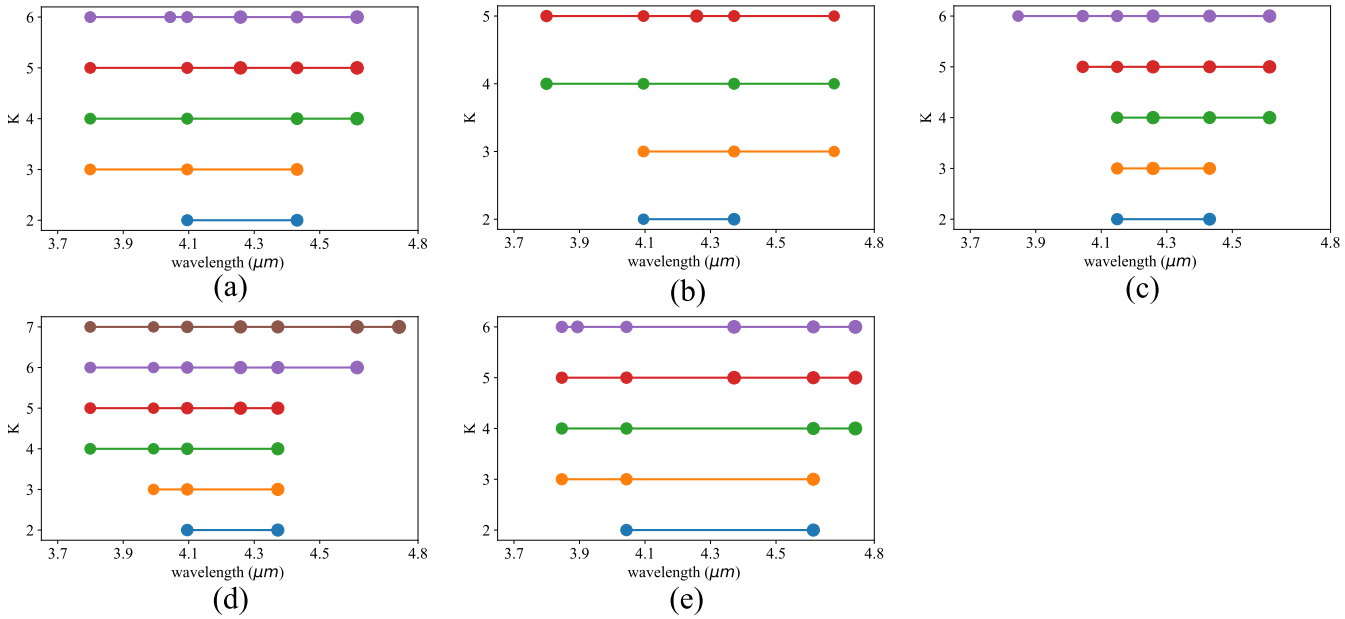


Fig. 11. Variation of spectral parameters selected by WBMARL with K. (a)–(e) Results for vegetation background, building background, road background, and sky background in Dataset 1, and airplane target in Dataset 2, respectively.

TABLE II
METHODS INFERENCE TIME COMPARISON

Methods	Inference time (s)
WBMARL	0.713
OIF+SCR	1257.012
ASPS+SCR	0.019
OPBS+SCR	0.176
AMSPSO	195.676
CFSP	1.411
ERLA	114.870
S4P	2.073
CNNeGA	791.760

S4P, OPBS+SCR, and ASPS+SCR demonstrated faster computational performance.

F. Analysis of the Sensitivity

To analyze the sensitivity of WBMARL to the number of parameter combinations, an experiment about the trend of the JM distance of the selected images changing with the number of selected parameters is conducted. In addition, the variation in spectral parameters selected with respect to K is investigated. It is evident that WBMARL exhibits distinct sequential decision-making characteristics. As the number of selected spectral parameters increases, new features are incrementally added to the existing selection. The model first identifies the most critical spectral parameters, then progressively supplements them until achieving the maximum JM distance. The variation of spectral parameters selected by WBMARL with K is shown in Fig. 11.

For the multiple sets of multispectral image data from Datasets 1 and 2, the variation in the resulting JM distance with changes in the number of selected spectral parameter combinations is shown in Fig. 12. For Dataset 1, the JM distance for the different backgrounds increases and then decreases with the increase of the number of spectral parameter combinations. Initially, as the number of spectral parameter combinations increases, the band combinations contain more information, and the spectral separability improves. However, once a certain point is exceeded, the newly added spectral parameter combinations provide less additional information or are highly correlated with the already selected bands, resulting in a decrease in spectral separability. For the vegetation background, the JM distance of the spectral parameter combinations selected by the WBMARL is 1.996 when the number is 5, which is greater than 1.956 and 1.981 when the number is 4 and 6, respectively. Meanwhile, for the road and sky backgrounds, the JM distance is the largest when the number of spectral parameter combinations is 5. For the building background, the JM distance reaches the maximum of 1.995 when the number of spectral parameter combinations is 4, indicating good spectral separability. Increasing the number of parameter combinations will increase the time consumption for target detection, so it is more appropriate to choose the smallest number of combinations while ensuring separability. The number of spectral parameter combinations determined by the JM distance is more applicable. The JM distance is greater than 1.99 for different backgrounds, which can be considered that there is good spectral separability. For Dataset 2, when the number of spectral parameter combinations is 5, the JM distance of the spectral parameter combinations selected by the WBMARL reaches 1.993, which is higher than the JM distances when the combination numbers are 2, 3, 4, and 6. It has the best spectral separability and is most conducive to target detection. The WBMARL outputs combinations of spectral parameters one by one. Therefore, when more parameters are selected, the parameters will include the selection results

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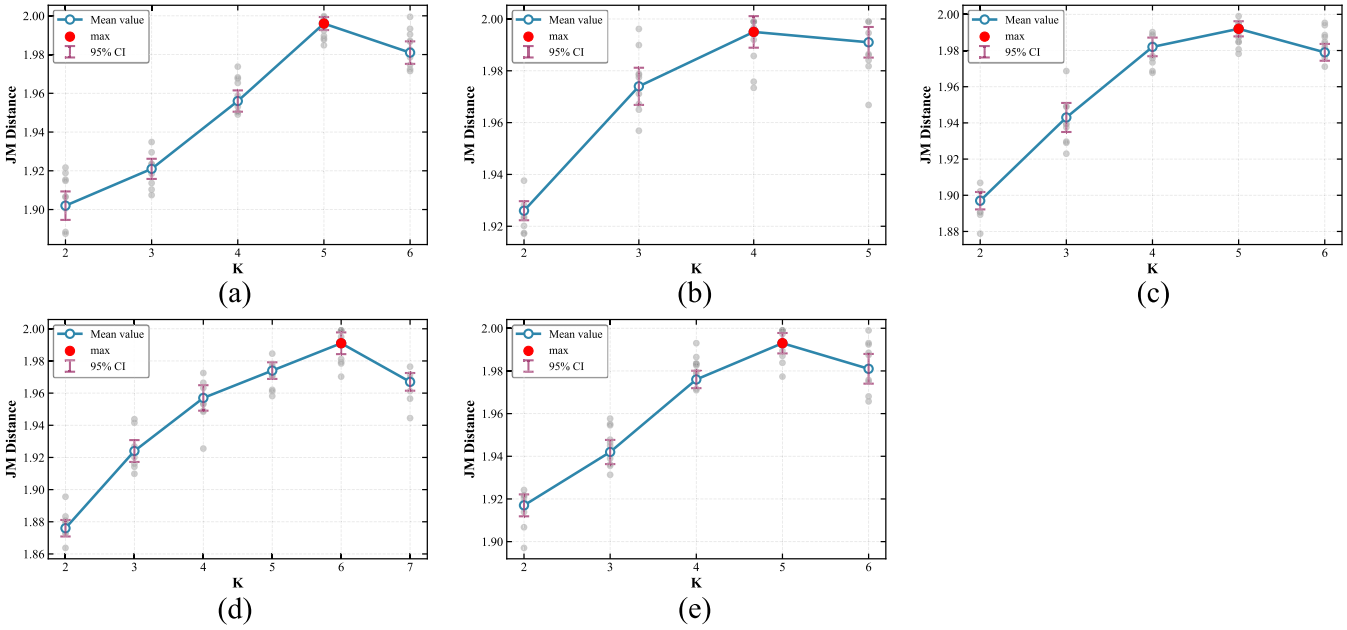


Fig. 12. Sensitivity of the WBMARL to the number of selected parameters. K represents the number of the selected spectral parameters, and the gray points represent the JM distances for different sets, the blue circles denote the mean values, the red points indicate the instances where the mean value of JM distance reaches its maximum, and the error bars represent the 95% confidence interval. (a)–(e) Results for vegetation background, building background, road background, and sky background in Dataset 1, and airplane target in Dataset 2, respectively.

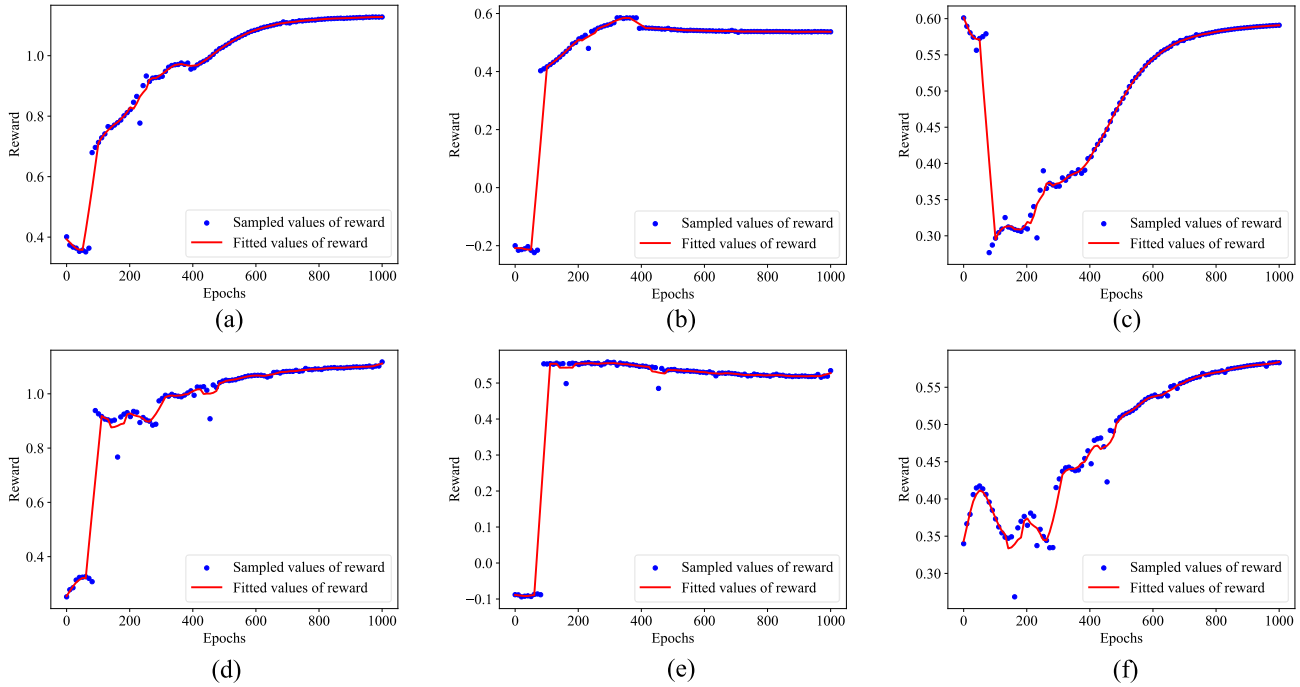


Fig. 13. Variation of the reward with training epochs. (a)–(c) Variation of R_t, R_1, and R_2 for Dataset 1, respectively. (d)–(f) Variation of R_t, R_1, and R_2 for Dataset 2.

when the number is smaller. It is quite obvious that for longer wavelengths, the selected bandwidths tend to be larger. For the simulated targets and simulated interferences in different backgrounds, some of the selected spectral parameters will be the same. Among them, the bands near 4.2 μm correspond to the emission characteristics of carbon dioxide gas, and there are significant differences between the targets and interferences.

G. Analysis of the Convergence

Convergence is critically important in multiagent systems. For multiagent deep reinforcement learning models, where deriving explicit mathematical expressions for convergence and stability proves challenging, the reward R_t of the WBMARL is analyzed. The rewards for the two agents are formulated to ensure continuity and boundedness, which

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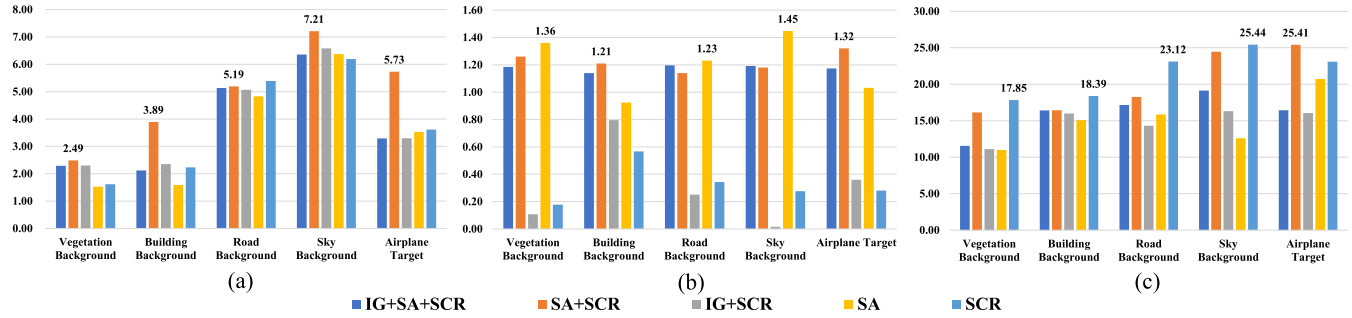


Fig. 14. Evaluation of different rewards for Datasets 1 and 2. (a)–(c) Results for MIE, SA, and SCR, respectively. IG+SA+SCR indicates that the reward consists of IG, SA, and SCR, where IG and SA share the same coefficient. SA+SCR denotes that the reward is composed of SA and SCR. IG+SCR means that the reward comprises IG and SCR. SA indicates that the reward consists solely of SA, and SCR signifies that the reward is composed only of SCR.

are necessary prerequisites for model convergence. The variation of the R_t with training epochs is shown in Fig. 13. For Dataset 1, the reward exhibits significant fluctuations in the initial training phase, indicating instability. However, as training progresses, the reward gradually stabilizes around a fixed value. After 600 training epochs, the reward remains nearly unchanged, demonstrating successful convergence. The reward for Dataset 2 shows slow growth early in training and stabilizes after approximately 500 epochs. The difference in the reward on Datasets 1 and 2 may be attributed to the varying complexity of the scenarios in the two datasets. The scenes in Dataset 1 are more complex, leading to a slower increase in the reward, which suggests that a longer learning time is required to adapt to such environments. Nevertheless, WBMARL still achieves high reward values after only a small number of training epochs, demonstrating its strong perceptual capability and adaptability to the data.

In addition to analyzing the reward R_t, examining R_1 and R_2 individually provides deeper insights into the performance of each agent during training. The results of the wavelength selection agent and the bandwidth selection agent significantly influence R_1 and R_2, respectively. For both Datasets 1 and 2, R_1 rapidly increases to a high value in the early training stage, indicating that the wavelength selection agent quickly identifies wavelength combinations with significant spectral difference. Meanwhile, R_2 grows slowly, suggesting that the bandwidth selection agent continuously explores bandwidth combinations to maximize the SCR. Notably, as the SCR of the selected spectral parameters combination increases, the spectral difference inevitably decreases, highlighting the coupled relationship between wavelength and bandwidth selections. Ultimately, during the convergence phase of the reward function, the optimal outcome is not achieved by solely maximizing SA or SCR, but rather by balancing both factors.

In addition, another noteworthy observation is that despite the distinct value ranges of the SCR and the SA, the agents balance the contributions of R_1 and R_2, resulting in comparable value domains for both rewards. This indicates that the two agents operate in a relative balance and maintain an effective collaborative mechanism.

H. Analysis of Different Rewards

The design of reward significantly impacts multiagent reinforcement learning models, as different reward functions can drive the model to develop corresponding selection preferences. Therefore, various reward functions are analyzed. WBMARL employs both SA and SCR, with their coefficients learned by the network. In addition, Wu et al. [39] suggested that information gain (IG) can serve as a reward function for band selection. Hence, IG is introduced as a reward for wavelength selection. As shown in Fig. 14, the comparison reveals that reward consists of SA and SCR maximizes MIE across all scenarios. For spectral differences, reward consists of SA and SCR, reward consists of SA alone, and reward consists of IG, SA, and SCR yield high values, indicating that SA significantly influences the spectral differences in selection outcomes. For SCR, the highest values are observed when reward is composed of SA and SCR or SCR alone, though overall differences among the results are relatively small. This may be because there is a consistent tendency to select bands with high image quality and high SCR, regardless of the rewards. The above results demonstrate that using SA or SCR alone as the reward is insufficient. Compared to rewards composed of IG and SCR, or IG, SA, and SCR, the reward combining SA and SCR delivers superior performance, exhibiting greater spectral difference and higher SCR. For WBMARL and the spectral parameter selection task, the introduction of IG does not yield significant benefits. Therefore, adopting a reward composed of SA and SCR is more appropriate.

I. Analysis of η and γ

The configuration of the learning rate η and discount factor γ significantly influences model performance.

A larger learning rate accelerates learning in the initial stages but leads to instability and difficulties in convergence, while a smaller learning rate promotes stability but slows learning progress and increases the risk of falling into local optima. By analyzing metrics such as MIE, SA, and SCR of selection from models trained with different learning rates, the impact of the learning rate on WBMARL is evaluated. As shown in Fig. 15, the optimal learning rate for MIE falls within the range of 0.001–0.00001. When the learning rate is set to 0.00001, the SA and SCR indicate that, due to the small learning rate, the model learns slowly and has not yet reached the optimum. Thus, SA remains relatively high while SCR is comparatively low, which is consistent with the convergence analysis of the model. As the learning rate increases to 0.0001,

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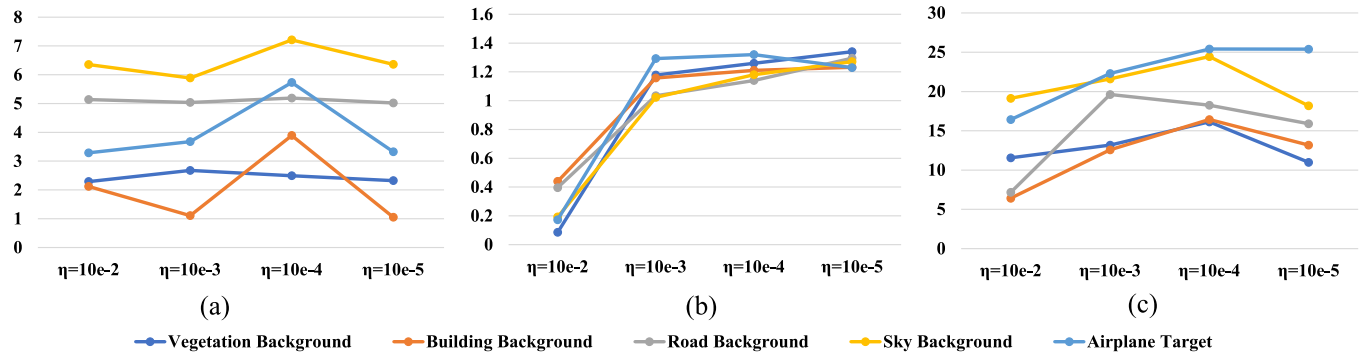


Fig. 15. Impact of learning rate η. (a)–(c) Results for MIE, SA, and SCR, respectively.

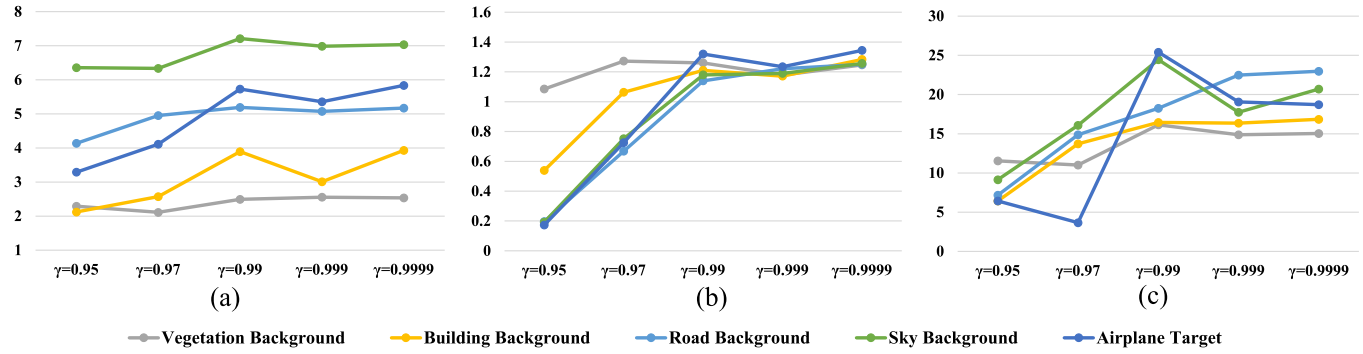


Fig. 16. Impact of discount factor γ. (a)–(c) Results for MIE, SA, and SCR, respectively.

faster learning enables the model to approach the optimal policy. In some scenarios, SA slightly decreases, but SCR increases significantly. Therefore, a learning rate of 0.0001 is ultimately selected.

Regarding the discount factor γ, a higher value encourages the model to prioritize long-term returns, while a lower value emphasizes short-term gains. However, both excessively large and small values can impair performance. As illustrated in Fig. 16, analysis shows that when the discount factor is set to 0.99, metrics such as MIE, SA, and SCR achieve optimal results in certain scenarios. As the discount factor decreases, SA declines significantly for building background, road background, sky background, and airplane target. As it increases, SA also decreases, with MIE and SCR dropping substantially in some scenarios. Therefore, the discount factor is set to 0.99.

V. Conclusion and Outlook

In this article, we first illustrated the importance of spectral parameters selection for infrared multispectral detection based on an AOTF system. Second, we introduced the band selection methods from two aspects: selection metrics and selection methods. Then, we proposed the WBMARL for wavelength and bandwidth selections based on multiagent deep reinforcement learning. We regarded the wavelength and bandwidth selection tasks as a Markov game process. According to the requirements of SA and SCR of the detection bands for aerial targets, we defined the state, action, reward, state transfer function, and discount factor of the multiagent system. We designed the WBMARL, composed of SFEM and two agents. The SFEM is used to extract features from multispectral images captured by an AOTF system to construct the environment. The two agents are, respectively, used for wavelength and bandwidth selections. Through the cooperation between the agents and their interaction with the environment, the optimal policy for wavelength and bandwidth selections is obtained. Finally, we used the MADDPG algorithm to train the model to achieve stable convergence. The experimental results show that the spectral parameters selected by WBMARL are more conducive to the detection of targets in complex backgrounds and anti-interference. Our work presents a novel paradigm for spectral parameters selection, offering a valuable reference for researchers in related fields. However, the current WBMARL framework has certain limitations. The adopted agent network structure is relatively straightforward and may struggle to handle more complex state spaces. In future work, the framework could be enhanced in several directions. For instance, more sophisticated interagent communication mechanisms could be introduced to explicitly model the physical coupling between parameters, such as wavelength and bandwidth. In addition, the framework could be extended to tackle joint optimization problems involving a broader range of spectral parameters.

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